MD ENAMUL HAQUE

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RESEARCH INTERESTS

Efficient and multimodal AI for biomedical imaging, healthcare, and intelligent systems, with a focus on attention-guided deep learning, multi-task learning, medical vision-language models, and the extension of multi-modal reasoning toward vision-language-action and embodied AI.

EDUCATION

United International University (UIU) **Dhaka-1212, Bangladesh**
Bachelor of Science in Computer Science and Engineering (May 2018 – October 2022)
CGPA: 3.94/4.00 *(Top 1%) [Summa Cum Laude]*
Final Year Project: *"MixAugmentor and RePair: A Text and Image Data Augmentation Model."*
Supervisor: Nahid Hossain, Assistant Professor, Dept. of CSE & UG Program Coordinator, UIU.

PUBLICATIONS

1. M. E. Haque, M. R. Ahmed, R. S. Nila and S. Islam, *"Human Monkeypox Disease Detection Using Deep Learning and Attention Mechanisms,"* 2022 25th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2022, pp. 1069-1073, doi: 10.1109/ICCIT57492.2022.10055870.

2. M. F. Utsho, **M. E. Haque** and R. Rahman, *"Biometric Finger Vein Classification using Finger Inferred Images with ResNet50 and Attention Mechanisms,"* 2024 27th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2024, pp. 2940-2945, doi: 10.1109/ICCIT64611.2024.11021806.

3. M. E. Haque, A. Das, M. N. Huda and M. R. Ahmed, *"Nile Tilapia Pathogens Detection Using CNN Model Infused with Efficient Channel Attention Mechanism,"* 2025 28th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2025, pp. 4747-4752, doi: 10.1109/ICCIT68739.2025.11491097.

4. Ahmed, B.M., **Haque, M.E.**, Huq, A.K.O. et al. *FoodBD: a polygon-annotated meal image dataset of Bangladeshi cuisines with visual and nutritional labels.* BMC Res Notes 19, 17 (2026). https://doi.org/10.1186/s13104-025-07583-8

5. Das, A., **Haque, M. E.**, Alam, M. N., Shanto, M. M. H., Huda, M. N., & Kabir, M. H. (2026). *"MSDA-NET: Multi-Scale Dual Attention Network for Monkeypox Disease Detection."* **Manuscript under review in IEEE Access.** [Research Timeline: September 2025 - July 2026]

6. Haque, M.E., Ahmed, B.M., Huq, A.K.O. et al. *"BanglaFoodNet: A Multi-Task Deep Learning Framework for Joint Food Instance Segmentation, Fine-Grained Classification, and Glycemic-Load-Aware Nutrition Estimation from Bangladeshi Meal Images."* **Under review in Elsevier: Computers in Biology and Medicine.** [Research Timeline: December 2025 - May 2026]

7. Haque, M. E., Debnath, P., Rahi, A. H., Islam, S. M. J., & Shatabda, S. *"Can LLMs Feel the Metaphor? A Step-Level Dataset and Benchmark for Figurative Emotion Reasoning in Bangla Poetry."* Manuscript in preparation. [Research Timeline: January 2026 - Present]

8. Maroya, F. Y., Das, S., Shihab, M. S. R., Bhuiyan, M. H. A. J., Mim, N. S., **Haque, M.E.**, & Huda, M. N. *"Automated Detection of Vascular Discontinuities in Coronary Angiography Using a Hybrid Deep Learning-Classical Vesselness Framework"* Manuscript under review in (ICCIT-2026). [Research Timeline: March 2026 - July 2026]

RESEARCH EXPERIENCE

Funded Project by IAR, UIU: Institute for Advanced Research Lab. **Dhaka, Bangladesh**
Research Assistant *(March 2023 – February 2026)*
Project: "Blood Glucose Prediction from Daily Food Intake and Activities" | **Grant ID:** UIU/IAR/01/2021/SE/11 | **Investigator:** Md. Benzir Ahmed, Assistant Professor, CSE, UIU | **Co-Investigators:** Dr. Mahmuda Naznin (CSE, BUET), Dr. Md. Eunus Ali (Monash University), and Dr. Mohammad Mehedy Masud (UAE University) | **Focus:** Machine Learning and Health Informatics for Non-invasive Diabetes Management

FoodBD: A Polygon-Annotated Meal Image Dataset of Bangladeshi Cuisines with Visual and Nutritional Labels. [March 2023 – November 2025]

- Developed FoodBD, a culturally diverse Bangladeshi food image dataset containing 3,523 smartphone-captured meal images across 67 food categories with polygon-based annotations.
- Provided expert-estimated nutritional information for 1,837 images, including carbohydrate, protein, fat, fiber, calorie, and glycemic load, supporting AI-driven dietary assessment.

BanglaFoodNet: A Multi-Task Deep Learning Framework for Joint Food Instance Segmentation, Fine-Grained Classification, and Glycemic-Load-Aware Nutrition Estimation [December 2025 – February 2026]

- Designed BanglaFoodNet, a multi-task framework combining Swin-Tiny, CBAM-FPN, and task-specific heads for food segmentation, fine-grained classification, and nutrition estimation.
- Proposed GL-aware Huber loss, cross-task gated nutrition regression, and uncertainty-based multi-task loss weighting for unified food analysis on FoodBD.

TEACHING EXPERIENCE

Dept. of Computer Science and Engineering, United International University — Dhaka, Bangladesh

Lecturer *(November 2025 – Present);* ***Instructor*** *(January 2024 – October 2025) (Promoted to Lecturer in November 2025.)*

- **Course conducted:** Taught 8 trimesters (Spring 2024 – Summer 2026), covering 13 distinct courses spanning core CS theory and systems, including Data Structures & Algorithms I/II, Database Management Systems, Operating Systems Laboratory, Computer Networks Laboratory, Object-Oriented Programming, Discrete Mathematics, Digital Logic Design, and System Analysis & Design.

Dept. of Computer Science and Technology, Educity-Bangladesh — Dhaka, Bangladesh

Instructor *(November 2022 – December 2023)*

- Delivered academic and industry-oriented instruction to Diploma-in-Engineering (Polytechnic) students at Bangladesh's first career-guidance ed-tech platform.
- **Course Conducted:** Data Structures and Algorithms, Database Management System, Programming in Java, Basic Python, Basic Application Development Using Python, Operating System, Principles of Software Engineering, Computer Peripherals and Interfacing, and Network Administration and Services.

Dept. of Computer Science and Engineering, United International University — Dhaka, Bangladesh

Undergraduate Assistant *(October 2021 – October 2022)*

- **Course Assisted:** Structured Programming Language Lab, Microprocessor, Micro-controller, and Interfacing Lab, Computer Networks Lab, Electronics Lab

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java
Deep Learning Frameworks: PyTorch, Keras, HuggingFace Transformers, Scikit-Learn
Computer Vision Libraries: OpenCV, PIL, torchvision
Data & Scientific Computing: NumPy, Pandas, Matplotlib
Research Tools & Infrastructure: Git/GitHub, LaTeX, CUDA-enabled GPU training, Google Colab, Kaggle
Core CS & Teaching-Relevant Areas: Data Structures & Algorithms, Database Management Systems (MySQL), Operating Systems, Computer Networks, Object-Oriented Programming (Java, JavaFX), Django, Digital Logic Design
Embedded/Hardware (Coursework): Atmega32 Microcontroller, Arduino, Proteus, Logisim

HONORS AND AWARDS

Summa Cum Laude Scholastic Award, UIU — February 2024
Best Presentation Award, Session-242, ICCIT 2022 — December 2022
Merit-based Tuition Fee Waiver (Top 1% of cohort), awarded every trimester, UIU — 2018–2022
1st Position, CSE Project Show, Software Project: Musicity (Online Music Streaming Platform), UIU — Spring 2021
3rd Position, CSE Project Show (Final Year Design Project: MixAugmentor and RePair), UIU - Summer 2022
9th Position of 37 teams, Intra-University AI Contest, UIU — June 2022

REFERENCES

- **Prof. Dr. Mohammad Nurul Huda**, Professor and Ex-Head of the Dept. of CSE, United International University, Email: mnh@cse.uiu.ac.bd, Contact: +880-1673476433
- **Md. Benzir Ahmed**, Assistant Professor, Dept. of CSE, United International University, Email: benzir@cse.uiu.ac.bd Contact: +880-1983792928